

LAB 9

1. Perform the following tasks:

a. Create Student table with following attributes (STUDENT_ID , FIRST_NAME, LAST_NAME, PHONE_NUMBER, MARKS, COURSE_ID).

OUERY:

```
mysql> use lab9_1;
```

OUTPUT:

```
mysql> use lab9_1;  
Database changed  
mysql>
```

OUERY:

```
CREATE TABLE student (  
    student_id INT(10) PRIMARY KEY NOT NULL,  
    first_name VARCHAR(40) NOT NULL,  
    last_name VARCHAR(40) NOT NULL,  
    phone_number VARCHAR(15) NOT NULL,  
    marks INT(4),  
    course_id INT(10) NOT NULL,  
    FOREIGN KEY (course_id) REFERENCES course(course_id));
```

OUTPUT:

```
mysql> CREATE TABLE student (  
->     student_id INT(10) PRIMARY KEY NOT NULL,  
->     first_name VARCHAR(40) NOT NULL,  
->     last_name VARCHAR(40) NOT NULL,  
->     phone_number VARCHAR(15) NOT NULL,  
->     marks INT(4),  
->     course_id INT(10) NOT NULL,  
->     FOREIGN KEY (course_id) REFERENCES course(course_id)  
-> );  
Query OK, 0 rows affected, 3 warnings (0.04 sec)
```

b. Create Course table with following attributes (COURSE_ID, COURSE_NAME).

QUERY:

```
CREATE TABLE course(  
    course_id int(10) PRIMARY KEY NOT NULL,  
    course_name varchar(100) NOT NULL
```

);

OUTPUT:

```
mysql> CREATE TABLE course(  
->     course_id int(10) PRIMARY KEY NOT NULL,  
->     course_name varchar(100) NOT NULL  
-> );  
Query OK, 0 rows affected, 1 warning (0.03 sec)
```

c. Write a SQL statement to insert 8 records with your own value into the tables.

ANSWER:

(1). Inserting into student table

INSERT INTO student (student_id, first_name, last_name, phone_number, marks, course_id)
VALUES

- (1, 'Ashi', 'Chouhan', '9876543210', 85, 1),
- (2, 'Aryan', 'Chouhan', '9123456789', 90, 2),
- (3, 'Suhani', 'Chouhan', '9988776655', 78, 3),
- (4, 'Nikhil', 'Singh', '9765432109', 88, 1),
- (5, 'Dinesh', 'Chaudhary', '9876512345', 92, 2),
- (6, 'Ranjesh', 'Kushwaha', '9234567890', 80, 3),
- (7, 'Riya', 'Yadav', '9887766554', 75, 4),
- (8, 'Roshani', 'Singh', '9321654987', 83, 5);

OUTPUT:

```
mysql> INSERT INTO student (student_id, first_name, last_name, phone_number,  
marks, course_id) VALUES  
-> (1, 'Ashi', 'Chouhan', '9876543210', 85, 1),  
-> (2, 'Aryan', 'Chouhan', '9123456789', 90, 2),  
-> (3, 'Suhani', 'Chouhan', '9988776655', 78, 3),  
-> (4, 'Nikhil', 'Singh', '9765432109', 88, 1),  
-> (5, 'Dinesh', 'Chaudhary', '9876512345', 92, 2),  
-> (6, 'Ranjesh', 'Kushwaha', '9234567890', 80, 3),  
-> (7, 'Riya', 'Yadav', '9887766554', 75, 4),  
-> (8, 'Roshani', 'Singh', '9321654987', 83, 5);  
Query OK, 8 rows affected (0.01 sec)  
Records: 8 Duplicates: 0 Warnings: 0
```

(2). Inserting into course table:

INSERT INTO course (course_id, course_name) VALUES

- (1, 'Mathematics'),
- (2, 'Physics'),

(3, 'Chemistry'),
(4, 'Biology'),
(5, 'Computer Science'),
(6, 'History'),
(7, 'Economics'),
(8, 'English Literature');

OUTPUT:

```
mysql> INSERT INTO course (course_id, course_name) VALUES
-> (1, 'Mathematics'),
-> (2, 'Physics'),
-> (3, 'Chemistry'),
-> (4, 'Biology'),
-> (5, 'Computer Science'),
-> (6, 'History'),
-> (7, 'Economics'),
-> (8, 'English Literature');
Query OK, 8 rows affected (0.01 sec)
Records: 8 Duplicates: 0 Warnings: 0
```

d. Write a query to get the number of students with the same course.

QUERY:

```
mysql> SELECT course_id, COUNT(student_id) AS number_of_students
-> FROM student
-> GROUP BY course_id;
```

OUTPUT:

```
mysql> SELECT course_id, COUNT(student_id) AS number_of_students
-> FROM student
-> GROUP BY course_id;
+-----+-----+
| course_id | number_of_students |
+-----+-----+
| 1 | 2 |
| 2 | 2 |
| 3 | 2 |
| 4 | 1 |
| 5 | 1 |
+-----+-----+
5 rows in set (0.00 sec)
```

f. Write a query to get the student name, course name and marks of the students.

QUERY:

```
mysql> SELECT s.first_name, s.last_name, c.course_name, s.marks
-> FROM student s
-> JOIN course c ON s.course_id = c.course_id;
```

OUTPUT:

```
mysql> SELECT s.first_name, s.last_name, c.course_name, s.marks
-> FROM student s
-> JOIN course c ON s.course_id = c.course_id;
```

first_name	last_name	course_name	marks
Ashi	Chouhan	Mathematics	85
Aryan	Chouhan	Physics	90
Suhani	Chouhan	Chemistry	78
Nikhil	Singh	Mathematics	88
Dinesh	Chaudhary	Physics	92
Ranjesh	Kushwaha	Chemistry	80
Riya	Yadav	Biology	75
Roshani	Singh	Computer Science	83

```
8 rows in set (0.00 sec)
```

g. Write a query to get the Average marks of students course wise.

QUERY:

```
mysql> SELECT c.course_name, AVG(s.marks) AS average_marks
-> FROM student s
-> JOIN course c ON s.course_id = c.course_id
-> GROUP BY c.course_name;
```

OUTPUT:

```
mysql> SELECT c.course_name, AVG(s.marks) AS average_marks
-> FROM student s
-> JOIN course c ON s.course_id = c.course_id
-> GROUP BY c.course_name;
```

course_name	average_marks
Mathematics	86.5000
Physics	91.0000
Chemistry	79.0000
Biology	75.0000
Computer Science	83.0000

```
5 rows in set (0.01 sec)
```

2. Create database for hospital management system & Perform the following tasks:

a. Create HEALTH CARE WORKERS table with following attributes (EMPLOYEE_ID , FIRST_NAME, LAST_NAME, EMAIL, PHONE_NUMBER, HIRE_DATE, SALARY, DESIGNATION).

QUERY:

```
CREATE DATABASE hospital_management_system;
```

OUTPUT:

```
mysql> CREATE DATABASE hospital_management_system;
Query OK, 1 row affected (0.01 sec)

mysql> USE hospital_management_system;
Database changed

mysql>
```

QUERY:

```
mysql> CREATE TABLE HEALTH_CARE_WORKERS (  
->  EMPLOYEE_ID INT PRIMARY KEY,  
->  FIRST_NAME VARCHAR(50),  
->  LAST_NAME VARCHAR(50),  
->  EMAIL VARCHAR(100),  
->  PHONE_NUMBER VARCHAR(15),  
->  HIRE_DATE DATE,  
->  SALARY DECIMAL(10, 2),  
->  DESIGNATION VARCHAR(50)  
-> );
```

OUTPUT:

```
mysql> CREATE TABLE HEALTH_CARE_WORKERS (  
->  EMPLOYEE_ID INT PRIMARY KEY,  
->  FIRST_NAME VARCHAR(50),  
->  LAST_NAME VARCHAR(50),  
->  EMAIL VARCHAR(100),  
->  PHONE_NUMBER VARCHAR(15),  
->  HIRE_DATE DATE,  
->  SALARY DECIMAL(10, 2),  
->  DESIGNATION VARCHAR(50)  
-> );  
Query OK, 0 rows affected (0.05 sec)
```

b. Create PATIENT table with following attributes (PATIENT_ID,NAME, PHONE_NUMBER).

QUERY:

```
mysql> CREATE TABLE PATIENT (  
->  PATIENT_ID INT PRIMARY KEY,  
->  NAME VARCHAR(100),  
->  PHONE_NUMBER VARCHAR(15)  
-> );
```

OUTPUT:

```
mysql> CREATE TABLE PATIENT (  
->  PATIENT_ID INT PRIMARY KEY,  
->  NAME VARCHAR(100),  
->  PHONE_NUMBER VARCHAR(15)  
-> );  
Query OK, 0 rows affected (0.03 sec)
```

c. Write a SQL statement to insert 10 records with your own value into the tables.

QUERY:

```
INSERT INTO HEALTH_CARE_WORKERS (EMPLOYEE_ID, FIRST_NAME, EMAIL,  
PHONE_NUMBER, HIRE_DATE, SALARY, DESIGNATION)
```

VALUES

```
(1, 'Amit', 'amit@gmail.com', '9876543210', '2022-01-15', 30000.00, 'Doctor'),  
(2, 'Sita', 'sita@gmail.com', '9123456789', '2021-11-20', 32000.00, 'Nurse'),  
(3, 'Rajesh', 'rajesh@gmail.com', '9988776655', '2023-04-10', 40000.00, 'Surgeon'),  
(4, 'Aarti', 'aarti@gmail.com', '9765432109', '2020-07-05', 45000.00, 'Doctor'),  
(5, 'Vijay', 'vijay@gmail.com', '9876512345', '2019-09-15', 22000.00, 'Technician'),  
(6, 'Neeta', 'neeta@gmail.com', '9234567890', '2021-03-12', 35000.00, 'Nurse'),  
(7, 'Ravi', 'ravi@gmail.com', '9887766554', '2022-08-25', 37000.00, 'Physician'),  
(8, 'Pooja', 'pooja@gmail.com', '9321654987', '2020-01-01', 27000.00, 'Technician'),  
(9, 'Arun', 'arun@gmail.com', '9998887777', '2018-05-30', 31000.00, 'Doctor'),  
(10, 'Kavita', 'kavita@gmail.com', '8887776666', '2023-02-10', 42000.00, 'Surgeon');
```

OUTPUT:

```
mysql> INSERT INTO HEALTH_CARE_WORKERS (EMPLOYEE_ID, FIRST_NAME, EMAIL, PHON  
E_NUMBER, HIRE_DATE, SALARY, DESIGNATION)  
-> VALUES  
-> (1, 'Amit', 'amit@gmail.com', '9876543210', '2022-01-15', 30000.00, '  
Doctor'),  
-> (2, 'Sita', 'sita@gmail.com', '9123456789', '2021-11-20', 32000.00, '  
Nurse'),  
-> (3, 'Rajesh', 'rajesh@gmail.com', '9988776655', '2023-04-10', 40000.0  
0, 'Surgeon'),  
-> (4, 'Aarti', 'aarti@gmail.com', '9765432109', '2020-07-05', 45000.00,  
'Doctor'),  
-> (5, 'Vijay', 'vijay@gmail.com', '9876512345', '2019-09-15', 22000.00,  
'Technician'),  
-> (6, 'Neeta', 'neeta@gmail.com', '9234567890', '2021-03-12', 35000.00,  
'Nurse'),  
-> (7, 'Ravi', 'ravi@gmail.com', '9887766554', '2022-08-25', 37000.00, '  
Physician'),  
-> (8, 'Pooja', 'pooja@gmail.com', '9321654987', '2020-01-01', 27000.00,  
'Technician'),  
-> (9, 'Arun', 'arun@gmail.com', '9998887777', '2018-05-30', 31000.00, '  
Doctor'),  
-> (10, 'Kavita', 'kavita@gmail.com', '8887776666', '2023-02-10', 42000.  
00, 'Surgeon');  
Query OK, 10 rows affected (0.02 sec)  
Records: 10 Duplicates: 0 Warnings: 0
```

QUERY:

```
INSERT INTO PATIENT (PATIENT_ID, NAME, PHONE_NUMBER)
```

VALUES

```
(1, 'Ashi', '9876543210'),
```

```
(2, 'Aryan', '9123456789'),
(3, 'Suhani', '9988776655'),
(4, 'Nikhil', '9765432109'),
(5, 'Dinesh', '9876512345'),
(6, 'Ranjesh', '9234567890'),
(7, 'Riya', '9887766554'),
(8, 'Roshani', '9321654987'),
(9, 'Meera', '9998887777'),
(10, 'Ravi', '8887776666');
```

OUTPUT:

```
mysql> INSERT INTO PATIENT (PATIENT_ID, NAME, PHONE_NUMBER)
-> VALUES
-> (1, 'Ashi', '9876543210'),
-> (2, 'Aryan', '9123456789'),
-> (3, 'Suhani', '9988776655'),
-> (4, 'Nikhil', '9765432109'),
-> (5, 'Dinesh', '9876512345'),
-> (6, 'Ranjesh', '9234567890'),
-> (7, 'Riya', '9887766554'),
-> (8, 'Roshani', '9321654987'),
-> (9, 'Meera', '9998887777'),
-> (10, 'Ravi', '8887776666');
Query OK, 10 rows affected (0.01 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

d. Write a query to get the names (first_name, last_name), Designation, salary.

QUERY:

```
mysql> SELECT FIRST_NAME, DESIGNATION, SALARY
-> FROM HEALTH_CARE_WORKERS;
```

OUTPUT:

```
mysql> SELECT FIRST_NAME, DESIGNATION, SALARY
-> FROM HEALTH_CARE_WORKERS;
+-----+-----+-----+
| FIRST_NAME | DESIGNATION | SALARY |
+-----+-----+-----+
| Amit      | Doctor      | 30000.00 |
| Sita      | Nurse       | 32000.00 |
| Rajesh    | Surgeon     | 40000.00 |
| Aarti     | Doctor      | 45000.00 |
| Vijay     | Technician  | 22000.00 |
| Neeta     | Nurse       | 35000.00 |
| Ravi      | Physician   | 37000.00 |
| Pooja     | Technician  | 27000.00 |
| Arun      | Doctor      | 31000.00 |
| Kavita    | Surgeon     | 42000.00 |
+-----+-----+-----+
10 rows in set (0.00 sec)
```

e. Write a query to get the number of employees with the same Designation

QUERY:

```
mysql> SELECT DESIGNATION, COUNT(*) AS NumberOfEmployees  
-> FROM HEALTH_CARE_WORKERS  
-> GROUP BY DESIGNATION;
```

OUTPUT:

```
mysql> SELECT DESIGNATION, COUNT(*) AS NumberOfEmployees  
-> FROM HEALTH_CARE_WORKERS  
-> GROUP BY DESIGNATION;  
+-----+-----+  
| DESIGNATION | NumberOfEmployees |  
+-----+-----+  
| Doctor      | 3 |  
| Nurse       | 2 |  
| Surgeon     | 2 |  
| Technician  | 2 |  
| Physician   | 1 |  
+-----+-----+  
5 rows in set (0.01 sec)
```

f. Write a query to get employee name who are getting salary more than 25000.

QUERY:

```
SELECT FIRST_NAME  
FROM HEALTH_CARE_WORKERS  
WHERE SALARY > 25000;
```

OUTPUT:

```
mysql> SELECT FIRST_NAME  
-> FROM HEALTH_CARE_WORKERS  
-> WHERE SALARY > 25000;  
+-----+  
| FIRST_NAME |  
+-----+  
| Amit       |  
| Sita       |  
| Rajesh     |  
| Aarti      |  
| Neeta      |  
| Ravi       |  
| Pooja      |  
| Arun       |  
| Kavita     |  
+-----+  
9 rows in set (0.01 sec)
```

g. Fetch HEALTH CARE WORKERS name using their employee id.

QUERY:

```
mysql> SELECT FIRST_NAME  
-> FROM HEALTH_CARE_WORKERS  
-> WHERE EMPLOYEE_ID = 5;
```

OUTPUT:

```
mysql> SELECT FIRST_NAME  
-> FROM HEALTH_CARE_WORKERS  
-> WHERE EMPLOYEE_ID = 5;  
+-----+  
| FIRST_NAME |  
+-----+  
| Vijay      |  
+-----+  
1 row in set (0.00 sec)
```

3.Consider two tables, customers and orders, with the following structures:

- a) Customers Table: customer_id (Primary Key) first_name Last_name
- b) Orders Table: order_id (Primary Key) customer_id (Foreign Key) order_date
Total_amount
- c) Write an SQL query to retrieve the first and last names of customers along with the order date and total amount of their orders.
- d) Use an INNER JOIN to connect the two tables.

SETTING UP THE TABLES:

```
mysql> CREATE DATABASE LAB9_3;
```

```
mysql> CREATE DATABASE LAB9_3;  
Query OK, 1 row affected (0.01 sec)
```

```
mysql> CREATE TABLE customers (  
-> customer_id INT PRIMARY KEY,  
-> first_name VARCHAR(50),  
-> last_name VARCHAR(50)  
-> );
```

```
mysql> CREATE TABLE customers (  
-> customer_id INT PRIMARY KEY,  
-> first_name VARCHAR(50),  
-> last_name VARCHAR(50)  
-> );  
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> INSERT INTO customers (customer_id, first_name, last_name)  
-> VALUES  
-> (1, 'Ashi', 'Chouhan'),
```

```
-> (2, 'Aryan', 'Chouhan'),  
-> (3, 'Suhani', 'Chouhan'),  
-> (4, 'Nikhil', 'Singh'),  
-> (5, 'Dinesh', 'Chaudhary'),  
-> (6, 'Ranjesh', 'Kushwaha'),  
-> (7, 'Riya', 'Yadav'),  
-> (8, 'Roshani', 'Singh');
```

```
mysql> INSERT INTO customers (customer_id, first_name, last_name)  
-> VALUES  
-> (1, 'Ashi', 'Chouhan'),  
-> (2, 'Aryan', 'Chouhan'),  
-> (3, 'Suhani', 'Chouhan'),  
-> (4, 'Nikhil', 'Singh'),  
-> (5, 'Dinesh', 'Chaudhary'),  
-> (6, 'Ranjesh', 'Kushwaha'),  
-> (7, 'Riya', 'Yadav'),  
-> (8, 'Roshani', 'Singh');  
Query OK, 8 rows affected (0.01 sec)  
Records: 8 Duplicates: 0 Warnings: 0
```

```
mysql> CREATE TABLE orders (  
-> order_id INT PRIMARY KEY,  
-> customer_id INT,  
-> order_date DATE,  
-> total_amount DECIMAL(10, 2),  
-> FOREIGN KEY (customer_id) REFERENCES customers(customer_id)  
-> );
```

```
mysql> CREATE TABLE orders (  
-> order_id INT PRIMARY KEY,  
-> customer_id INT,  
-> order_date DATE,  
-> total_amount DECIMAL(10, 2),  
-> FOREIGN KEY (customer_id) REFERENCES customers(customer_id)  
-> );  
Query OK, 0 rows affected (0.06 sec)
```

```
mysql> INSERT INTO orders (order_id, customer_id, order_date, total_amount)  
-> VALUES  
-> (101, 1, '2024-08-01', 250.75),  
-> (102, 1, '2024-08-15', 150.00),  
-> (103, 2, '2024-08-20', 320.50),  
-> (104, 3, '2024-08-25', 180.00),  
-> (105, 4, '2024-08-30', 275.20),  
-> (106, 5, '2024-08-10', 200.00),  
-> (107, 6, '2024-08-12', 150.00),  
-> (108, 7, '2024-08-18', 120.00),  
-> (109, 8, '2024-08-22', 300.00);
```

```
mysql> INSERT INTO orders (order_id, customer_id, order_date, total_amount)
-> VALUES
-> (101, 1, '2024-08-01', 250.75),
-> (102, 1, '2024-08-15', 150.00),
-> (103, 2, '2024-08-20', 320.50),
-> (104, 3, '2024-08-25', 180.00),
-> (105, 4, '2024-08-30', 275.20),
-> (106, 5, '2024-08-10', 200.00),
-> (107, 6, '2024-08-12', 150.00),
-> (108, 7, '2024-08-18', 120.00),
-> (109, 8, '2024-08-22', 300.00);
Query OK, 9 rows affected (0.01 sec)
Records: 9 Duplicates: 0 Warnings: 0
```

CUSTOMERS TABLE:

```
mysql> SELECT * FROM CUSTOMERS;
+-----+-----+-----+
| customer_id | first_name | last_name |
+-----+-----+-----+
| 1 | Ashi | Chouhan |
| 2 | Aryan | Chouhan |
| 3 | Suhani | Chouhan |
| 4 | Nikhil | Singh |
| 5 | Dinesh | Chaudhary |
| 6 | Ranjesh | Kushwaha |
| 7 | Riya | Yadav |
| 8 | Roshani | Singh |
+-----+-----+-----+
8 rows in set (0.00 sec)
```

ORDERS TABLE:

```
mysql> SELECT * FROM ORDERS;
+-----+-----+-----+-----+
| order_id | customer_id | order_date | total_amount |
+-----+-----+-----+-----+
| 101 | 1 | 2024-08-01 | 250.75 |
| 102 | 1 | 2024-08-15 | 150.00 |
| 103 | 2 | 2024-08-20 | 320.50 |
| 104 | 3 | 2024-08-25 | 180.00 |
| 105 | 4 | 2024-08-30 | 275.20 |
| 106 | 5 | 2024-08-10 | 200.00 |
| 107 | 6 | 2024-08-12 | 150.00 |
| 108 | 7 | 2024-08-18 | 120.00 |
| 109 | 8 | 2024-08-22 | 300.00 |
+-----+-----+-----+-----+
9 rows in set (0.00 sec)
```

Now:

QUERY:

```
mysql> SELECT c.first_name, c.last_name, o.order_date, o.total_amount
-> FROM customers c
-> INNER JOIN orders o ON c.customer_id = o.customer_id;
```

OUTPUT:

```
mysql> SELECT c.first_name, c.last_name, o.order_date, o.total_amount
-> FROM customers c
-> INNER JOIN orders o ON c.customer_id = o.customer_id;
+-----+-----+-----+-----+
| first_name | last_name | order_date | total_amount |
+-----+-----+-----+-----+
| Ashi | Chouhan | 2024-08-01 | 250.75 |
| Ashi | Chouhan | 2024-08-15 | 150.00 |
| Aryan | Chouhan | 2024-08-20 | 320.50 |
| Suhani | Chouhan | 2024-08-25 | 180.00 |
| Nikhil | Singh | 2024-08-30 | 275.20 |
| Dinesh | Chaudhary | 2024-08-10 | 200.00 |
| Ranjesh | Kushwaha | 2024-08-12 | 150.00 |
| Riya | Yadav | 2024-08-18 | 120.00 |
| Roshani | Singh | 2024-08-22 | 300.00 |
+-----+-----+-----+-----+
9 rows in set (0.00 sec)
```

4. Consider two tables, departments and employees, with the following structures:

Departments Table: department_id (Primary Key) department_name

Employees Table: employee_id (Primary Key) first_name last_name department_id (Foreign Key)

Write an SQL query to retrieve a list of all departments and the names of employees who belong to each department. Use a LEFT JOIN to include departments that have no employees.

SETTING UP THE TABLES:

```
mysql> CREATE DATABASE LAB9_4;
Query OK, 1 row affected (0.01 sec)

mysql> USE LAB9_4;
Database changed
```

```
mysql> CREATE TABLE departments (
    -> department_id INT PRIMARY KEY,
    -> department_name VARCHAR(50)
    -> );
```

```
mysql> CREATE TABLE departments (
    -> department_id INT PRIMARY KEY,
    -> department_name VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> INSERT INTO departments (department_id, department_name)
    -> VALUES
    -> (1, 'HR'),
    -> (2, 'Finance'),
    -> (3, 'IT'),
    -> (4, 'Marketing');
```

```
mysql> INSERT INTO departments (department_id, department_name)
-> VALUES
-> (1, 'HR'),
-> (2, 'Finance'),
-> (3, 'IT'),
-> (4, 'Marketing');
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0
```

```
mysql> CREATE TABLE employees (
->   employee_id INT PRIMARY KEY,
->   first_name VARCHAR(50),
->   last_name VARCHAR(50),
->   department_id INT,
->   FOREIGN KEY (department_id) REFERENCES departments(department_id)
-> );
```

```
mysql> CREATE TABLE employees (
->   employee_id INT PRIMARY KEY,
->   first_name VARCHAR(50),
->   last_name VARCHAR(50),
->   department_id INT,
->   FOREIGN KEY (department_id) REFERENCES departments(department_id)
-> );
Query OK, 0 rows affected (0.06 sec)
```

```
mysql> INSERT INTO employees (employee_id, first_name, last_name, department_id)
-> VALUES
-> (1, 'Ashi', 'Chouhan', 1),
-> (2, 'Aryan', 'Chouhan', 2),
-> (3, 'Suhani', 'Chouhan', 3),
-> (4, 'Nikhil', 'Singh', 1),
-> (5, 'Dinesh', 'Chaudhary', 2),
-> (6, 'Ranjesh', 'Kushwaha', NULL),
-> (7, 'Riya', 'Yadav', 4),
-> (8, 'Roshani', 'Singh', 3);
```

```
mysql> INSERT INTO employees (employee_id, first_name, last_name, department_id)
-> VALUES
-> (1, 'Ashi', 'Chouhan', 1),
-> (2, 'Aryan', 'Chouhan', 2),
-> (3, 'Suhani', 'Chouhan', 3),
-> (4, 'Nikhil', 'Singh', 1),
-> (5, 'Dinesh', 'Chaudhary', 2),
-> (6, 'Ranjesh', 'Kushwaha', NULL),
-> (7, 'Riya', 'Yadav', 4),
-> (8, 'Roshani', 'Singh', 3);
Query OK, 8 rows affected (0.01 sec)
Records: 8 Duplicates: 0 Warnings: 0
```

DEPARTMENTS TABLE:

```
mysql> SELECT * FROM DEPARTMENTS;
+-----+-----+
| department_id | department_name |
+-----+-----+
| 1 | HR |
| 2 | Finance |
| 3 | IT |
| 4 | Marketing |
+-----+-----+
4 rows in set (0.00 sec)
```

EMPLOYEES TABLE:

```
mysql> SELECT * FROM EMPLOYEES;
+-----+-----+-----+-----+
| employee_id | first_name | last_name | department_id |
+-----+-----+-----+-----+
| 1 | Ashi | Chouhan | 1 |
| 2 | Aryan | Chouhan | 2 |
| 3 | Suhani | Chouhan | 3 |
| 4 | Nikhil | Singh | 1 |
| 5 | Dinesh | Chaudhary | 2 |
| 6 | Ranjesh | Kushwaha | NULL |
| 7 | Riya | Yadav | 4 |
| 8 | Roshani | Singh | 3 |
+-----+-----+-----+-----+
8 rows in set (0.00 sec)
```

NOW,

QUERY:

```
mysql> SELECT d.department_name, e.first_name, e.last_name
-> FROM departments d
-> LEFT JOIN employees e ON d.department_id = e.department_id;
```

OUTPUT:

```
mysql> SELECT d.department_name, e.first_name, e.last_name
-> FROM departments d
-> LEFT JOIN employees e ON d.department_id = e.department_id;
+-----+-----+-----+
| department_name | first_name | last_name |
+-----+-----+-----+
| HR              | Ashi      | Chouhan   |
| HR              | Nikhil    | Singh     |
| Finance         | Aryan     | Chouhan   |
| Finance         | Dinesh    | Chaudhary |
| IT              | Suhani    | Chouhan   |
| IT              | Roshani   | Singh     |
| Marketing       | Riya      | Yadav     |
+-----+-----+-----+
7 rows in set (0.01 sec)
```

4. Write a program to show JDBC connection with MYSQL and perform the following operations:

Create table Customer with following fields:

Custno, Custame,Custaddress,Phoneno, City, Pincode, Country

Insert 5 records in Customer table.

- Insert values
- Delete values
- update city name Shimla to Shilong.
- Show table in the console

PROGRAM:

// ConnectJDBC

```
package LAB9_JDBC;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;

public class ConnectJDBC {

    public static Connection getConnection() throws SQLException {
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            return
DriverManager.getConnection("jdbc:mysql://localhost:3306/lab9_jdbc", "ashi",
"123456");
        } catch (ClassNotFoundException e) {
            e.printStackTrace();
            throw new SQLException("MySQL JDBC Driver not found.");
        }
    }
}
```

// DBOperations.java

```
package LAB9_JDBC;

import java.sql.Connection;
import java.sql.ResultSet;
import java.sql.SQLException;
import java.sql.Statement;

public class DBOperations {
    public static void main(String[] args) {
        Connection conn = null;
        Statement stmt = null;
        ResultSet rs = null;

        try {
            // Establish a connection to the database
            conn = ConnectJDBC.getConnection();
            stmt = conn.createStatement();

            // Create the Customer table if it doesn't exist
            String createTableSQL = "CREATE TABLE IF NOT EXISTS Customer (" +
                                    "Custno INT AUTO_INCREMENT PRIMARY KEY, " +
                                    "Custname VARCHAR(100), " +
                                    "Custaddress VARCHAR(255), " +
                                    "Phoneno VARCHAR(15), " +
                                    "City VARCHAR(50), " +
                                    "Pincode VARCHAR(10), " +
                                    "Country VARCHAR(50))";
            stmt.execute(createTableSQL); // Execute table creation
            System.out.println("Customer table created.");

            // Insert three records into the Customer table
            String insertSQL1 = "INSERT INTO Customer (Custname, Custaddress,
Phoneno, City, Pincode, Country) " +
                                "VALUES ('Ashi Chouhan', '12 MG Road',
'9876543210', 'Indore', '452001', 'India')";
            stmt.executeUpdate(insertSQL1); // Execute the insert operation

            String insertSQL2 = "INSERT INTO Customer (Custname, Custaddress,
Phoneno, City, Pincode, Country) " +
                                "VALUES ('Aryan Chouhan', '34 Brigade Road',
'9123456789', 'Bengaluru', '560001', 'India')";
            stmt.executeUpdate(insertSQL2); // Execute the insert operation

            String insertSQL3 = "INSERT INTO Customer (Custname, Custaddress,
Phoneno, City, Pincode, Country) " +
                                "VALUES ('Suhani Chouhan', '56 Marina Beach Road',
'9988776655', 'Chennai', '600002', 'India')";
            stmt.executeUpdate(insertSQL3); // Execute the insert operation

            System.out.println("3 records inserted into Customer table.");

            // Retrieve and display all records from the Customer table
            String selectSQL = "SELECT * FROM Customer";
            rs = stmt.executeQuery(selectSQL);

            while (rs.next()) {
                int custno = rs.getInt("Custno");
```

```

        String custname = rs.getString("Custname");
        String custaddress = rs.getString("Custaddress");
        String phoneno = rs.getString("Phoneno");
        String city = rs.getString("City");
        String pincode = rs.getString("Pincode");
        String country = rs.getString("Country");

        System.out.println("Custno: " + custno + ", Custname: " + custname
+ ", Custaddress: " + custaddress +
                                ", Phoneno: " + phoneno + ", City: " + city +
", Pincode: " + pincode + ", Country: " + country);
    }
    } catch (SQLException e) {
        e.printStackTrace();
    } finally {
        // Close resources
        try {
            if (rs != null) rs.close();
            if (stmt != null) stmt.close();
            if (conn != null) conn.close();
        } catch (SQLException e) {
            e.printStackTrace();
        }
    }
}
}
}

```

OUTPUT:

```

<terminated> DBOperations [Java Application] C:\Users\Ashi chouhan\p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_22.0.1.v20240426-1149\jre\bin\javaw.exe (3 Sept 2
Customer table created.
3 records inserted into Customer table.
Custno: 1, Custname: Ashi Chouhan, Custaddress: 12 MG Road, Phoneno: 9876543210, City: Indore, Pincode: 452001, Country: India
Custno: 2, Custname: Aryan Chouhan, Custaddress: 34 Brigade Road, Phoneno: 9123456789, City: Bengaluru, Pincode: 560001, Country: India
Custno: 3, Custname: Suhani Chouhan, Custaddress: 56 Marina Beach Road, Phoneno: 9988776655, City: Chennai, Pincode: 600002, Country: India
|

```